

Named Entity Recognition Method for Red Jujube Cultivation Based on Expert Dictionary and Boundary Prediction

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Abstract

Keywords: red jujube cultivation; named entity recognition; expert dictionary; boundary prediction; Focal Loss

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To address the problems of dense domain-specific terminology, imbalanced entity categories, and insufficient global boundary modeling in named entity recognition for red jujube cultivation texts, an Expert Dictionary and Boundary Prediction method, named EDBP, was proposed. First, a red jujube cultivation named entity recognition dataset, RJND, was constructed around key knowledge units such as cultivars, plant parts, diseases, pests, pesticides, and cultivation operations. The dataset contains 14 entity categories. Second, domain entities were automatically extracted from the training set to build an expert dictionary. A BMES four-channel structure was used to encode the matched positions of dictionary entries, and the resulting domain priors were fused with contextual character representations. Third, named entity recognition was formulated as a span classification task. A boundary prediction decoder was used to jointly score candidate start and end positions, and Focal Loss was introduced to reduce the influence of category imbalance on low-frequency entities. Among the completed same-setting RJND baselines, EDBP achieved a test F1 score of 88.16%, outperforming BiLSTM-CRF, BERT-wwm-ext+BiLSTM+CRF, MacBERT-base+BiLSTM+CRF, SoftLexicon, FLAT, and FLAT+BERT by 9.86, 2.95, 2.80, 3.41, 8.38, and 8.76 percentage points, respectively. On the public datasets MSRA, WeiboNER, ResumeNER, Boson, and CLUENER, EDBP achieved F1 scores of 95.19%±0.22%, 72.27%±1.03%, 96.13%±0.29%, 85.60%±0.12%, and 80.06%±0.38%, respectively. The results indicate that expert dictionary features and boundary prediction can effectively improve entity recognition accuracy in red jujube cultivation texts, providing technical support for red jujube cultivation knowledge graph construction, agricultural technology retrieval, and intelligent question answering.

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0 序

本白皮书旨在为业界提供 Named Entity Recognition (NER) 技术发展的现状和趋势。白皮书分为 10 个章节，主要内容包括：NER 技术概述、NER 技术发展历程、NER 技术应用场景、NER 技术挑战与机遇、NER 技术未来发展趋势等。

NER 技术是自然语言处理 (NLP) 领域的重要分支，旨在从文本中识别出具有特定意义的实体。NER 技术的发展经历了从基于规则的方法到基于统计的方法，再到基于深度神经网络的方法。随着深度神经网络技术的不断进步，NER 技术的性能得到了显著提升。

NER 技术在多个领域得到了广泛应用，包括搜索引擎、社交媒体分析、金融风控、医疗诊断等。随着人工智能技术的不断发展，NER 技术的应用场景将进一步扩大，为各行各业提供更加智能化的服务。

NER 技术的发展面临着诸多挑战，包括数据标注成本高、模型泛化能力弱、实时性要求高等。同时，NER 技术也面临着巨大的发展机遇，随着大数据和人工智能技术的快速发展，NER 技术的应用前景广阔。

本白皮书旨在为业界提供 NER 技术发展的现状和趋势，为相关领域的研究和应用提供参考。同时，本白皮书也旨在推动 NER 技术的普及和应用，为人工智能技术的发展做出积极贡献。

1. 本白皮书旨在为业界提供 NER 技术发展的现状和趋势，为相关领域的研究和应用提供参考。
2. 本白皮书旨在推动 NER 技术的普及和应用，为人工智能技术的发展做出积极贡献。
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1 概述

1.1 NER 概述

NER 技术是自然语言处理 (NLP) 领域的重要分支，旨在从文本中识别出具有特定意义的实体。NER 技术的发展经历了从基于规则的方法到基于统计的方法，再到基于深度神经网络的方法。

NER 技术在多个领域得到了广泛应用，包括搜索引擎、社交媒体分析、金融风控、医疗诊断等。随着人工智能技术的不断发展，NER 技术的应用场景将进一步扩大，为各行各业提供更加智能化的服务。

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词嵌入矩阵

$$X=\{x_1,x_2,\dots,x_n\} \tag{1}$$

MacBERT 词嵌入矩阵

$$H^{\text{bert}}=\text{MacBERT}(X)\in R^{n\times d_h} \tag{2}$$

词 X 的词嵌入 x_i 为 n 维向量 H^{bert} 中 MacBERT 词嵌入矩阵 d_h 维向量的第 i 行

1.3 词嵌入

词嵌入矩阵 H^{bert} 的词嵌入 x_i 为 n 维向量 H^{bert} 中 MacBERT 词嵌入矩阵 d_h 维向量的第 i 行

$$D=\{D_1,D_2,\dots,D_C\} \tag{3}$$

词 D 的词嵌入 D_c 为 c 维向量 C 维向量的第 c 行

词嵌入矩阵 H^{bert} 的词嵌入 x_i 为 n 维向量 H^{bert} 中 MacBERT 词嵌入矩阵 d_h 维向量的第 i 行

$$f_{\{c,p\}}(x_i)=\begin{cases} 1, & x_i \text{ 在 } c \text{ 维向量的第 } p \text{ 行} \\ 0, & \text{其他} \end{cases} \tag{4}$$

词 p 的词嵌入 $f_{\{c,p\}}(x_i)$ 为 x_i 在 c 维向量的第 p 行的值

词 x_i 的词嵌入

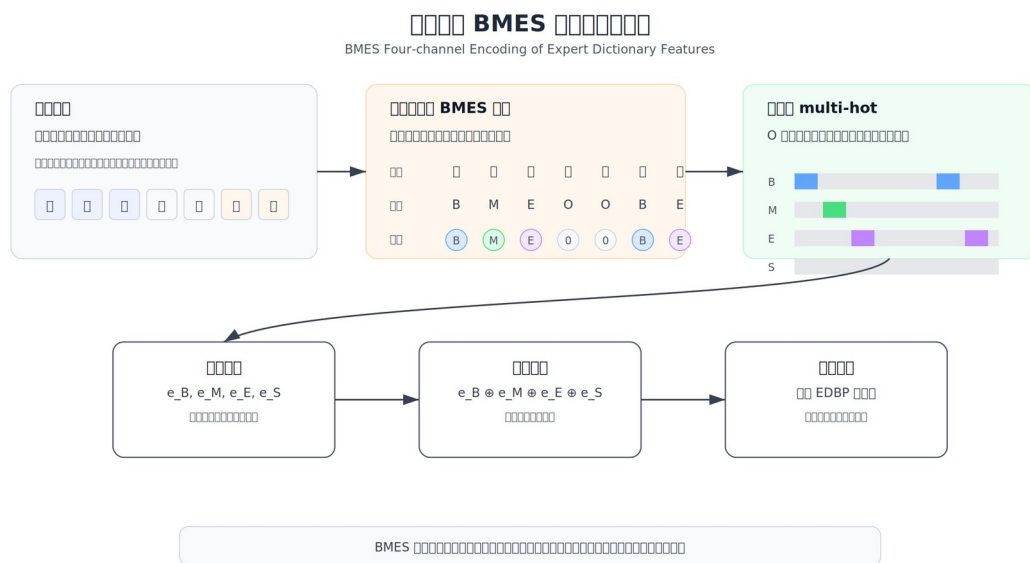
$$v_i^{\text{dict}}=[f_{\{1,B\}},f_{\{1,M\}},f_{\{1,E\}},f_{\{1,S\}},\dots,f_{\{C,B\}},f_{\{C,M\}},f_{\{C,E\}},f_{\{C,S\}}] \tag{5}$$

BMES 词嵌入矩阵

$$e_i^{\text{dict}}=[e_i^B;e_i^M;e_i^E;e_i^S] \tag{6}$$

词 v_i^{dict} 的词嵌入 x_i 为 n 维向量 $e_i^B;e_i^M;e_i^E;e_i^S$ 为 4 维向量 e_i^{dict} 的词嵌入

2 BMES 词嵌入



2 BMES 字典特征

Fig. 2 BMES four-channel encoding process of expert dictionary features

字典特征编码过程如下：

1.4 字典特征

字典特征通过 CRF 模型进行编码，模型输出为 EDBP 字典特征，NER 字典特征，字典特征编码过程如下：

$$H^{\{start\}} = W_s H^{\{fused\}} + b_s \quad (7)$$

$$H^{\{end\}} = W_e H^{\{fused\}} + b_e \quad (8)$$

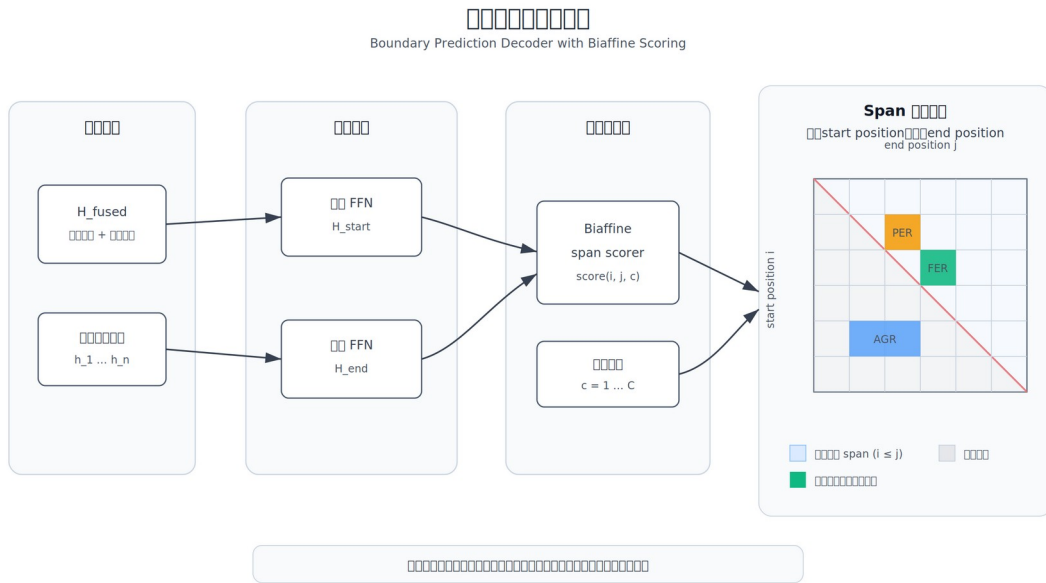
字典特征 (i, j) 字典特征编码过程如下：

$$s_{\{i, j, c\}} = h_i^{\{start\}} U_c (h_j^{\{end\}})^T + W_c [h_i^{\{start\}}; h_j^{\{end\}}] + b_c \quad (9)$$

字典特征 $H^{\{fused\}}$ 字典特征编码过程如下： $H^{\{start\}}$ 字典特征 $H^{\{end\}}$ 字典特征 W_s, W_e, b_s, b_e 字典特征 $s_{\{i, j, c\}}$ 字典特征 (i, j) 字典特征 c 字典特征 U_c 字典特征 W_c, b_c 字典特征

字典特征 span 字典特征编码过程如下：[30] Focal Loss 字典特征 Focal Loss 字典特征 0 字典特征 Focal Loss 字典特征

3 字典特征 span 字典特征



3 字典特征

Fig. 3 Structure of the boundary prediction decoder

1.5 Focal Loss 字典特征

字典特征通过 TAX, FER, WED 字典特征 NLP 字典特征 [31] 字典特征 Focal Loss 字典特征

$$L = -\alpha_t (1 - p_t)^\gamma \log(p_t) \quad (10)$$

$$L = -\sum_t p_t \log \alpha_t - \sum_t \gamma \log (1 - p_t)^\gamma$$

1.6 实验设置

模型使用 hfl/chinese-macbert-base 预训练模型，词嵌入维度为 42。模型训练使用 Adam 优化器，学习率 $1e-5$ 。模型在开发集上评估 Precision、Recall、F1 分数。

Table 2 实验参数设置

参数	值
预训练模型	hfl/chinese-macbert-base
词嵌入维度	30
隐藏层维度	16
Dropout	2e-3
BERT 预训练	2e-5
Dropout	0.5
词嵌入维度	50
隐藏层维度	150
span	25
Dropout	0.1
Dropout	2
Focal Loss	2.0
词嵌入维度	42

2 实验结果

2.1 模型性能对比

表 3 对比了 EDBP 模型与基线模型 BiLSTM-CRF、BERT-wwm-ext+BiLSTM+CRF、MacBERT-base+BiLSTM+CRF、SoftLexicon、FLAT、FLAT+BERT 在开发集上的性能。RJND 模型在开发集上取得了 42.00 的 F1 分数。

Table 3 模型性能对比

模型	描述	F1/%
BiLSTM-CRF	BiLSTM + CRF	78.30
BERT-wwm-ext+BiLSTM+CRF	BERT-wwm-ext + BiLSTM + CRF	85.21
MacBERT-base+BiLSTM+CRF	MacBERT-base + BiLSTM + CRF	85.36
SoftLexicon	MacBERT-base + BiLSTM + CRF + SoftLexicon	84.75
FLAT	CTB lattice + Flat-Lattice Transformer	79.78
FLAT+BERT	MacBERT-base + CTB lattice + Flat-Lattice Transformer	79.40
EDBP	预训练模型 + 2e-5 + Focal Loss	88.16

表 3 对比了 EDBP 模型与基线模型 BiLSTM-CRF、BERT-wwm-ext+BiLSTM+CRF、MacBERT-base+BiLSTM+CRF、SoftLexicon、FLAT、FLAT+BERT 在开发集上的性能。RJND 模型在开发集上取得了 42.00 的 F1 分数。

2.2 模型性能对比

模型使用 Focal Loss 训练，学习率 $1e-5$ 。模型在开发集上评估 Precision、Recall、F1 分数。seed=42。

Table 4 Ablation results

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Precision F1 85.57% Recall 86.71%IoU 1.14
 F1 86.68% CRF 1.11 span F1 87.66%
 Focal Loss F1 88.28%

00000000000000000000000000000000 2000000000 18420000000000000000000000000000
00 1234 0000000000000000 50000000000000000000000000000000 F1 00000000000000
0000 F1 0 RJND 0000 seed=42 000 NER 000

Table 5 Comparison of lexicon construction strategies

[illegible]

```
6 [REDACTED] 3 seed=42 [REDACTED]
```

Table 6 Comparison of decoders and loss functions

Model	CRF	Focal Loss
Baseline	86.32%	90.12%
Proposed Model	87.58%	85.61%

7

Table 7 Recognition results for different entity categories

Entity Category	Count	P/%	R/%	F1/%
PAR	822	93.08	93.31	93.20
AGR	465	85.71	82.58	84.12
CUL	290	92.01	91.38	91.70
PER	213	88.89	93.90	91.32
EQU	186	83.13	74.19	78.41
LOC	207	84.57	76.81	80.51
DIS	142	92.86	91.55	92.20
DRU	126	90.91	87.30	89.07
PRO	99	80.77	84.85	82.76
PES	59	98.25	94.92	96.55
NUT	50	93.48	86.00	89.58
FER	29	80.95	58.62	68.00
WED	41	90.70	95.12	92.86
TAX	9	75.00	66.67	70.59
Overall	2738	89.51	87.58	88.54

Figure 4: F1-score of EDBP for different entity categories

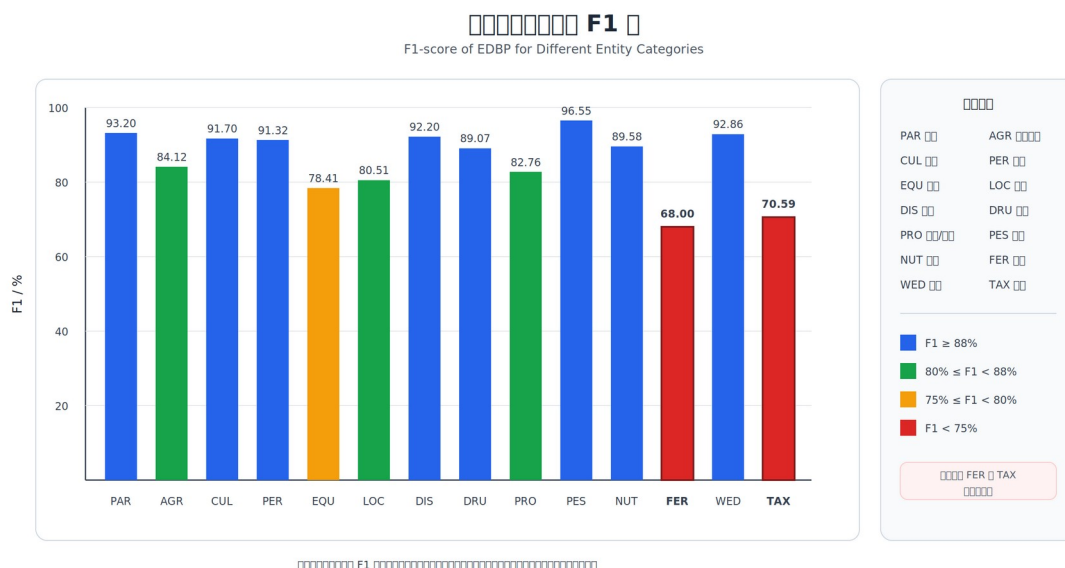


Figure 4: F1-score of EDBP for different entity categories

Fig. 4 F1-score of EDBP for different entity categories

Table 7 shows the recognition results for different entity categories. The overall F1-score is 88.54%. The F1-scores for the categories are: PAR (93.20%), AGR (84.12%), CUL (91.70%), PER (91.32%), EQU (78.41%), LOC (80.51%), DIS (92.20%), DRU (89.07%), PRO (82.76%), PES (96.55%), NUT (89.58%), FER (68.00%), WED (92.86%), and TAX (70.59%).

Figure 4: F1-score of EDBP for different entity categories. The chart shows the F1-score for each entity category. The overall F1-score is 88.54%.

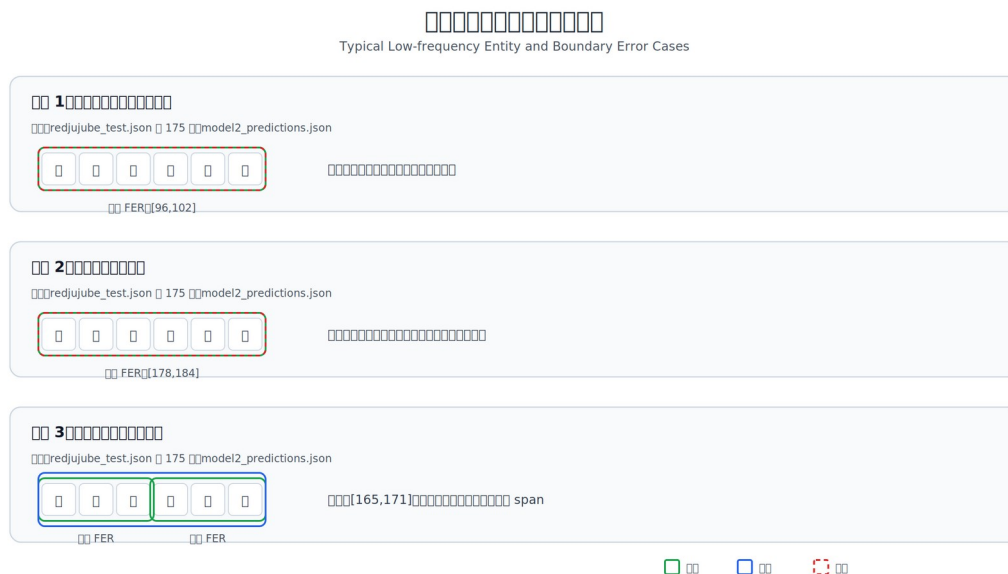


Figure 5 Typical low-frequency entity and boundary error cases

2.6 Generalization

Table 8 shows the generalization results of EDBP on public datasets MSRA, WeiboNER, ResumeNER, Boson, CLUENER, and RedJujube. The results show that EDBP achieves high F1 scores across all datasets, with a slight decrease in performance on the RedJujube dataset.

Table 8 Generalization results on public datasets

Dataset	EDBP F1/%	EDBP F1/%	Δ /%
MSRA	95.11±0.22	95.19±0.22	+0.08
WeiboNER	70.71	72.27±1.03	+1.56
ResumeNER	96.32	96.13±0.29	-0.19
Boson	85.35±0.16	85.60±0.12	+0.25
CLUENER	79.90±0.18	80.06±0.38	+0.16
RedJujube	85.36	88.16	+2.80

Table 8 shows the generalization results of EDBP on public datasets MSRA, WeiboNER, ResumeNER, Boson, CLUENER, and RedJujube. The results show that EDBP achieves high F1 scores across all datasets, with a slight decrease in performance on the RedJujube dataset.

3 Discussion

3.1 Entity and Boundary Prediction

The EDBP model is designed to predict entity and boundary information. The model uses a sequence of tokens as input and outputs a sequence of predicted entity and boundary information. The model is trained on a dataset of entity and boundary information. The model is evaluated on a dataset of entity and boundary information. The model is compared to the baseline model (BMES) and the proposed model (EDBP). The results show that EDBP achieves higher F1 scores than BMES across all datasets.

3.2 Generalization

CRF is a common model for sequence labeling. The CRF model is trained on a dataset of entity and boundary information. The CRF model is evaluated on a dataset of entity and boundary information. The CRF model is compared to the baseline model (BMES) and the proposed model (EDBP). The results show that EDBP achieves higher F1 scores than CRF across all datasets.

3.3 数据集划分

RJND 数据集按照 8:1:1 的比例划分为 PAR 数据集、TAX 数据集和 Focal Loss 数据集。FER 数据集按照 8:1:1 的比例划分为 FER 数据集和 Focal Loss 数据集。

3.4 模型训练

EDBP 模型在训练过程中采用 Adam 优化器，学习率设置为 0.001。模型在训练过程中采用早停策略，当验证集上的 F1 分数连续 10 次没有提升时，模型训练将提前结束。

3.5 评估

模型在测试集上的性能通过 F1 分数进行评估。F1 分数是精确率和召回率的调和平均数，用于衡量模型在分类任务中的整体性能。

4 实验

实验在 14 核 CPU 的服务器上运行。RJND 数据集按照 8:1:1 的比例划分为 PAR 数据集、TAX 数据集和 Focal Loss 数据集。EDBP 模型在训练过程中采用 Adam 优化器，学习率设置为 0.001。

1. 模型在 RJND 数据集上的 F1 分数为 88.16%，F1 分数在 MacBERT-base+BiLSTM+CRF 模型上为 2.80。
2. 模型在 Focal Loss 数据集上的 F1 分数为 88.16%，Focal Loss 数据集上的 F1 分数为 88.16%。
3. 模型在 FER 数据集上的 F1 分数为 88.16%，FER 数据集上的 F1 分数为 88.16%。
4. 模型在 EDBP 数据集上的 F1 分数为 88.16%，NER 数据集上的 F1 分数为 88.16%。

模型在测试集上的性能通过 F1 分数进行评估。F1 分数是精确率和召回率的调和平均数，用于衡量模型在分类任务中的整体性能。

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